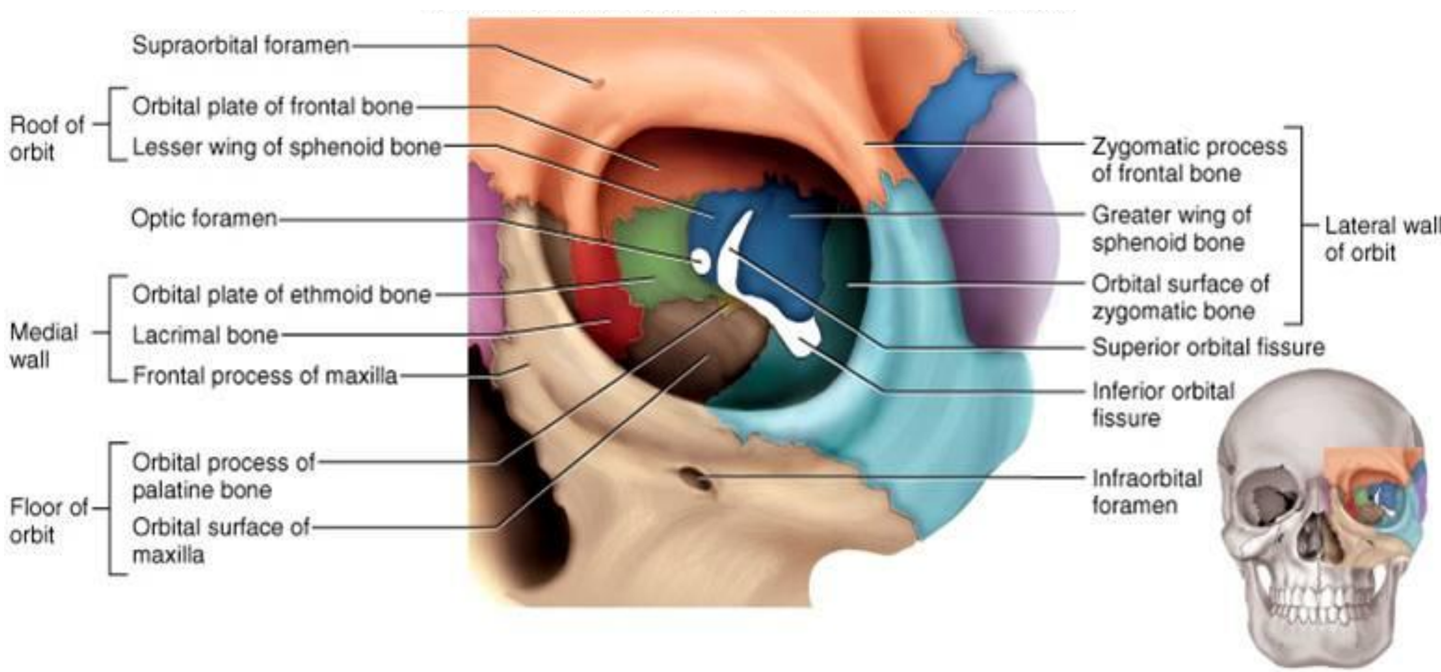
Orbital Cavity

(boundaries and extraocular muscles)

THE ORBIT

- Orbit is a bony cavity shaped like a four-sided pyramid
- Lying on each side of the root of the nose
- Lodges the eyeball.



Orbital fascia

Orbital fascia is the periosteum of the orbit which at the back is continuous with the duramater and sheath of the optic nerve.

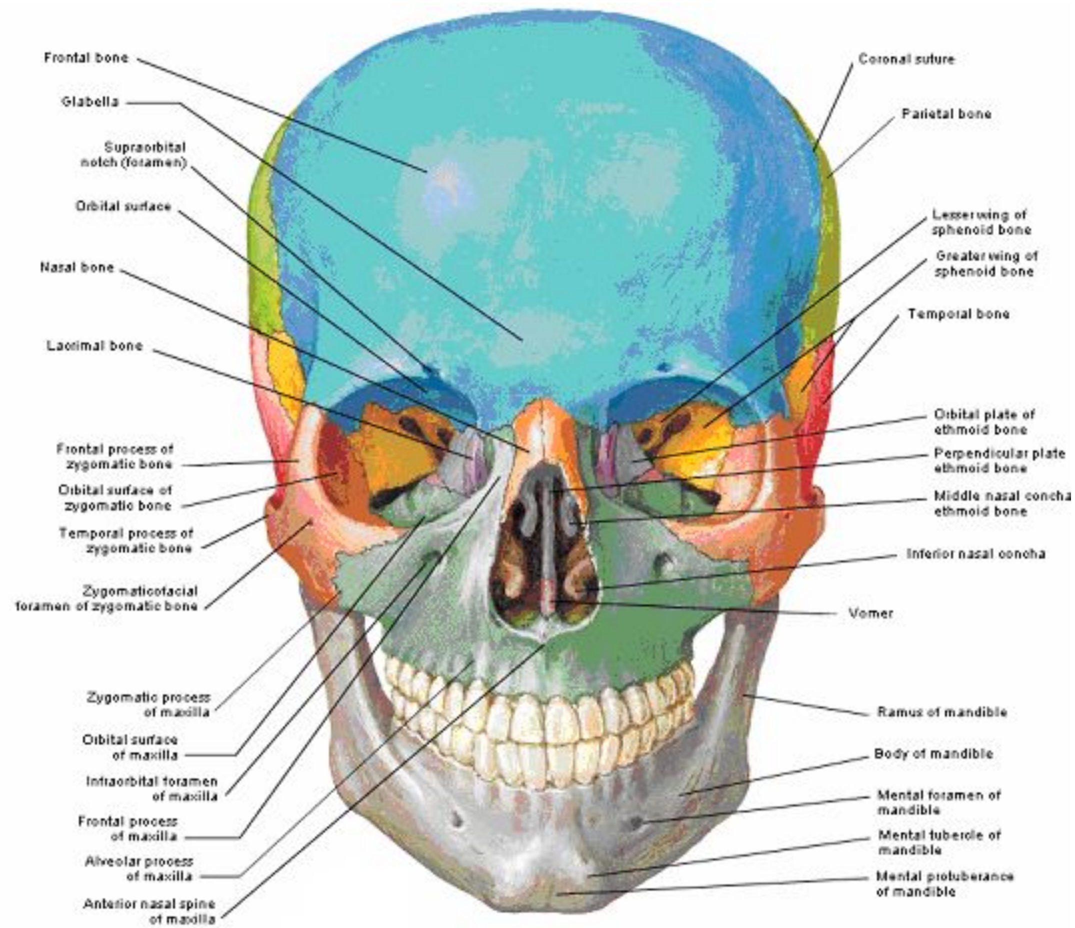
RELATIONS

SUPERIORLY

- Ant. Cranial fossa
- Meninges
- Frontal lobe of the cerebral hemisphere

MEDIALLY:

- Nasal cavity
- Ethmoidal sinus
- Sphenoidal sinus



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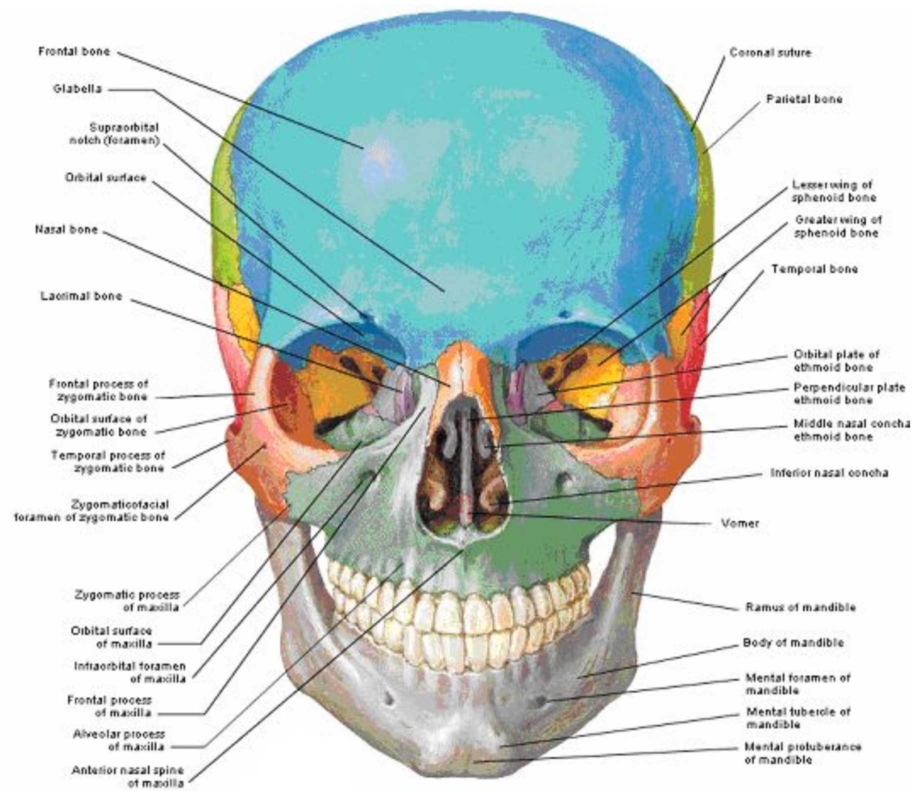
INFERIORLY:

- Maxillary sinus.

POSTEROLATERALLY:

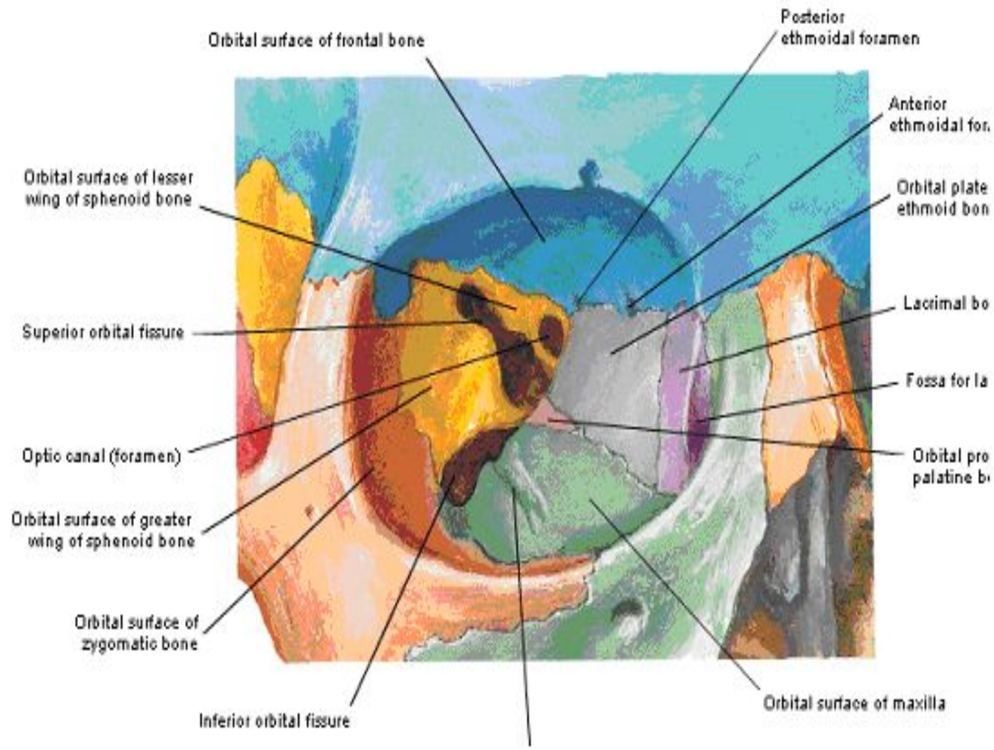
- Infratemporal fossa.
- Middle temporal fossa.



Boundaries

The orbit has

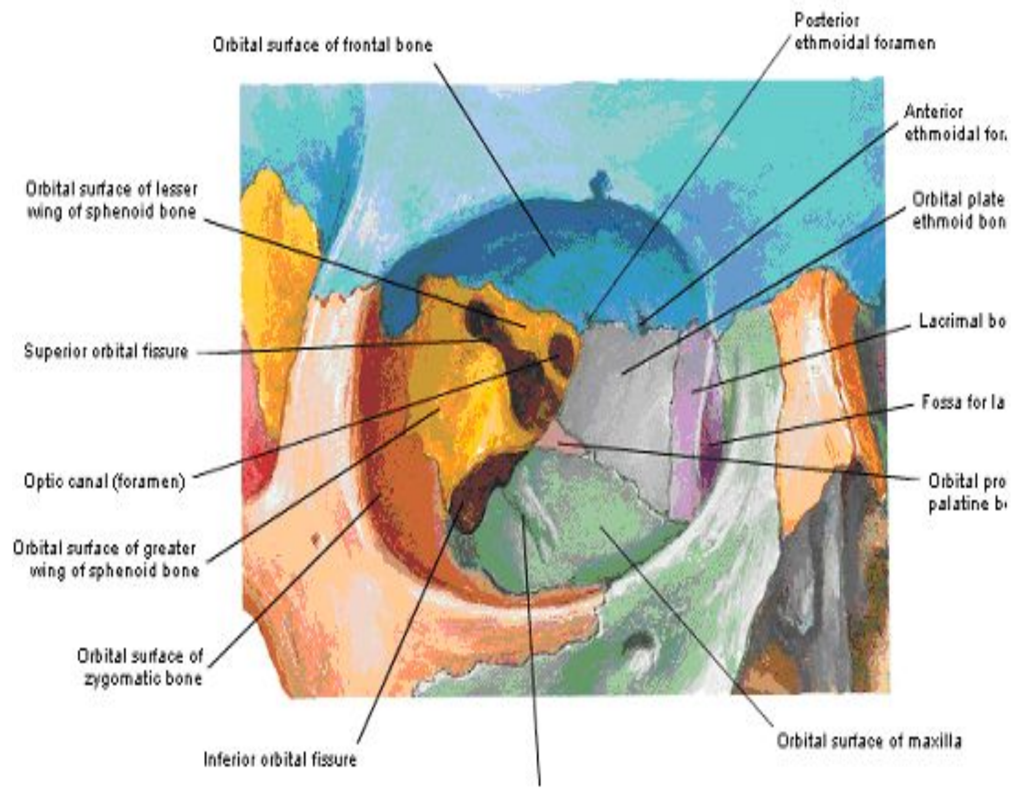
- **An apex** …..at the posterior end
- **A base**…….orbital margins
- **4 Walls**….Roof, Floor, Lateral & Medial walls



ROOF

Formed by

- Orbital part of the frontal bone
- Lesser wing of the sphenoid bone
- The roof separates the orbit from the anterior cranial fossa.



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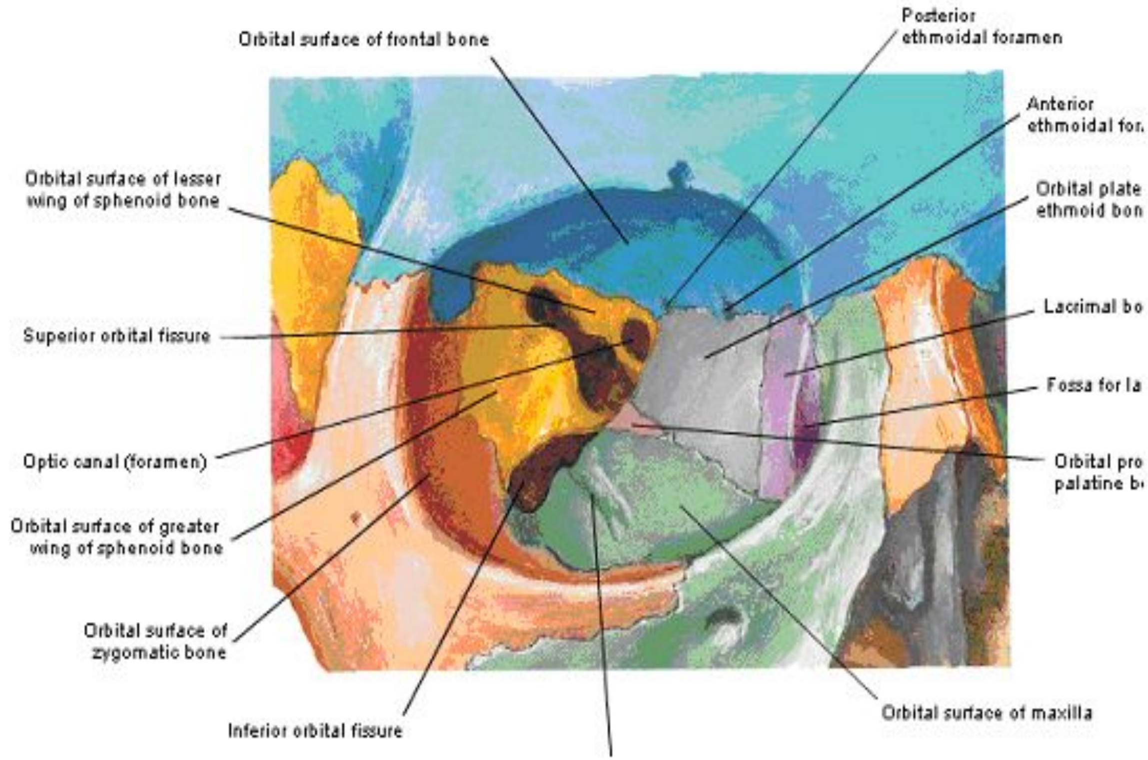
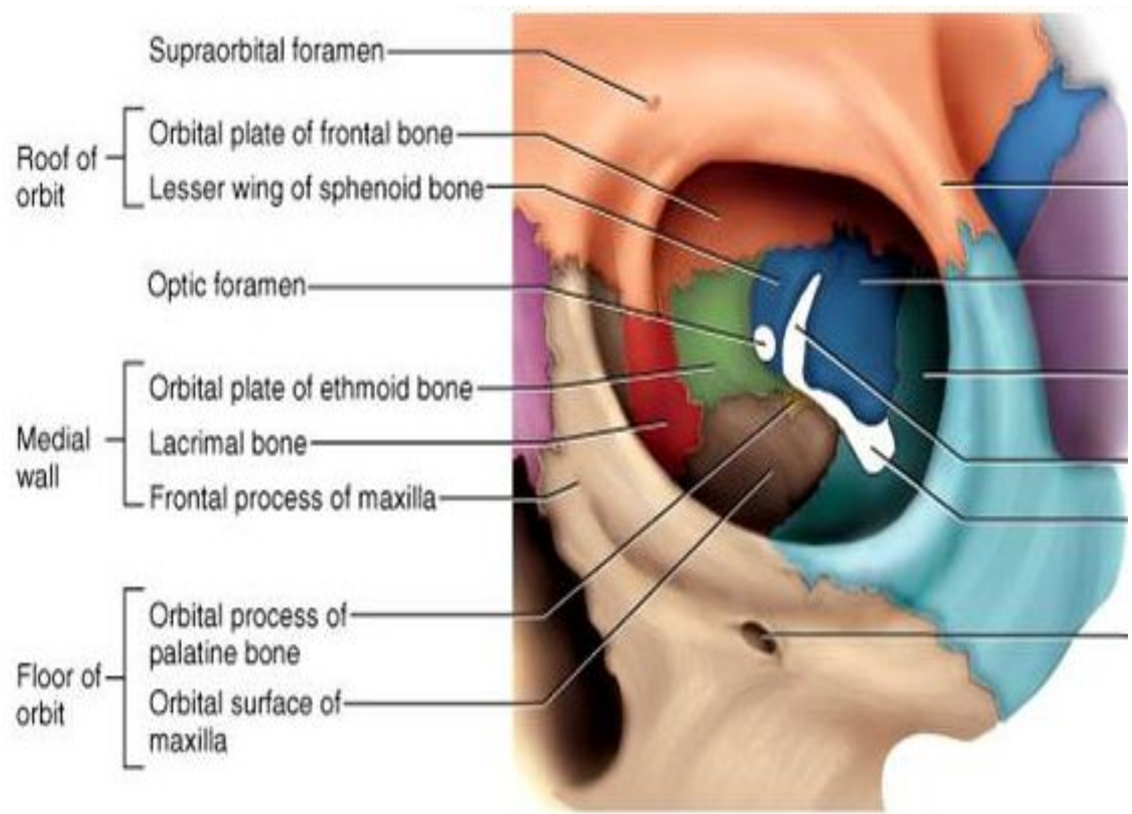
FEATURES:

- Lacrimal Fossa
- Optic Canal
- Trochlear fossa



MEDIAL WALL

- 5 cm long.
- Separates orbit from the ethmoidal ear cells.
- Formed by (before backwards).
- Frontal process of the maxilla.
- Lacrimal bone
- Orbital plate of the ethmoid bone.
- Body of the sphenoid bone.



Features

- Fossa for the Lacrimal sac….b/w ant. and post. Lacrimal crest
- Anterior Ethmoidal foramina…….24mm behind the ant.lacrimal crest
- Posterior ethmoidal foramina…….12mm behind this
- Optic nerve emerges 6mm further back

LATERAL WALL

- Thickest wall
- 5cm long

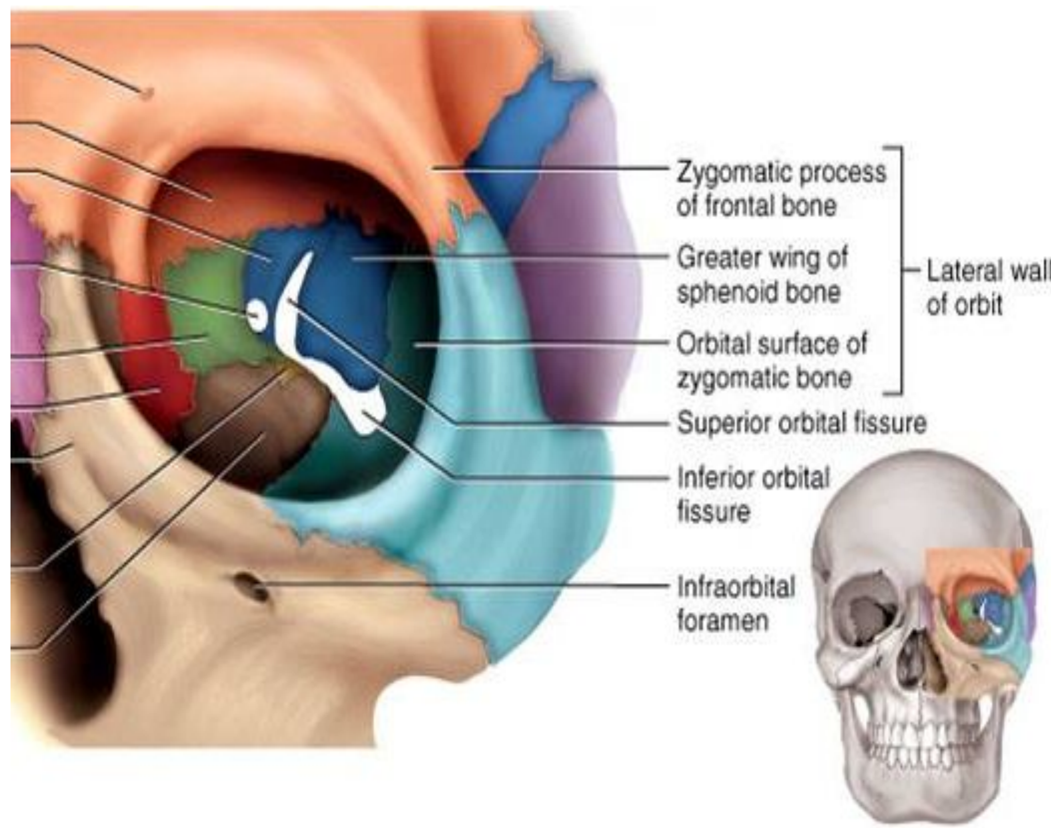
Formed :

- Anteriorly by orbital suface of the frontal process of the zygomatic bone

(Separates orbit from temporal fossa)

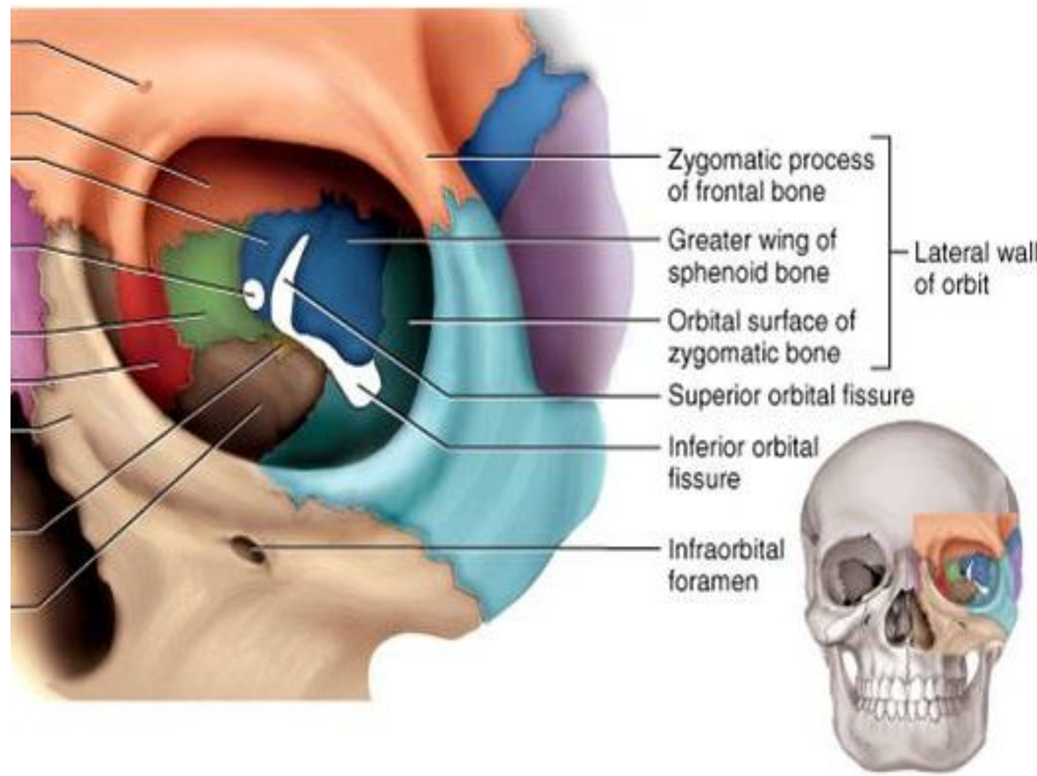
- Posteriorly by ant. Surface of the greater wing of the sphenoid bone

(Separates orbit from middle cranial fossa



Features

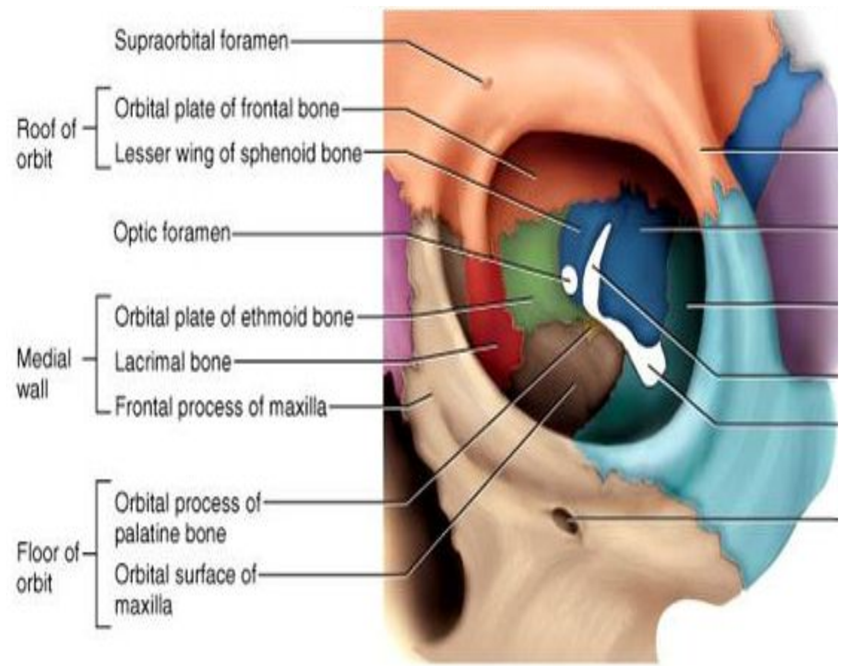
- **Superior orbital fissure**…gap b/w lateral wall and roof (leads to middle cranial fossa)
- **Inferior orbital fissure**…..gap b/w lateral wall & floor (leads to pterygopalatine & infratemporal fossae)



FLOOR

Formed by:

- Orbital surface of the maxilla
- Lower part of the orbital surfaces of the zygomatic bone
- Orbital process of the palatine bone
- Separates the orbit from the maxillary sinus



ORBITAL MARGINS

- Supraorbital Margin…frontal bone
- Infraorbital Margin….. zygomatic bone & maxilla
- Medial Margin……Ant. Lacrimal crest
- Frontal bone
- Lateral Margin…….. Frontal bone
- Zygomatic bone

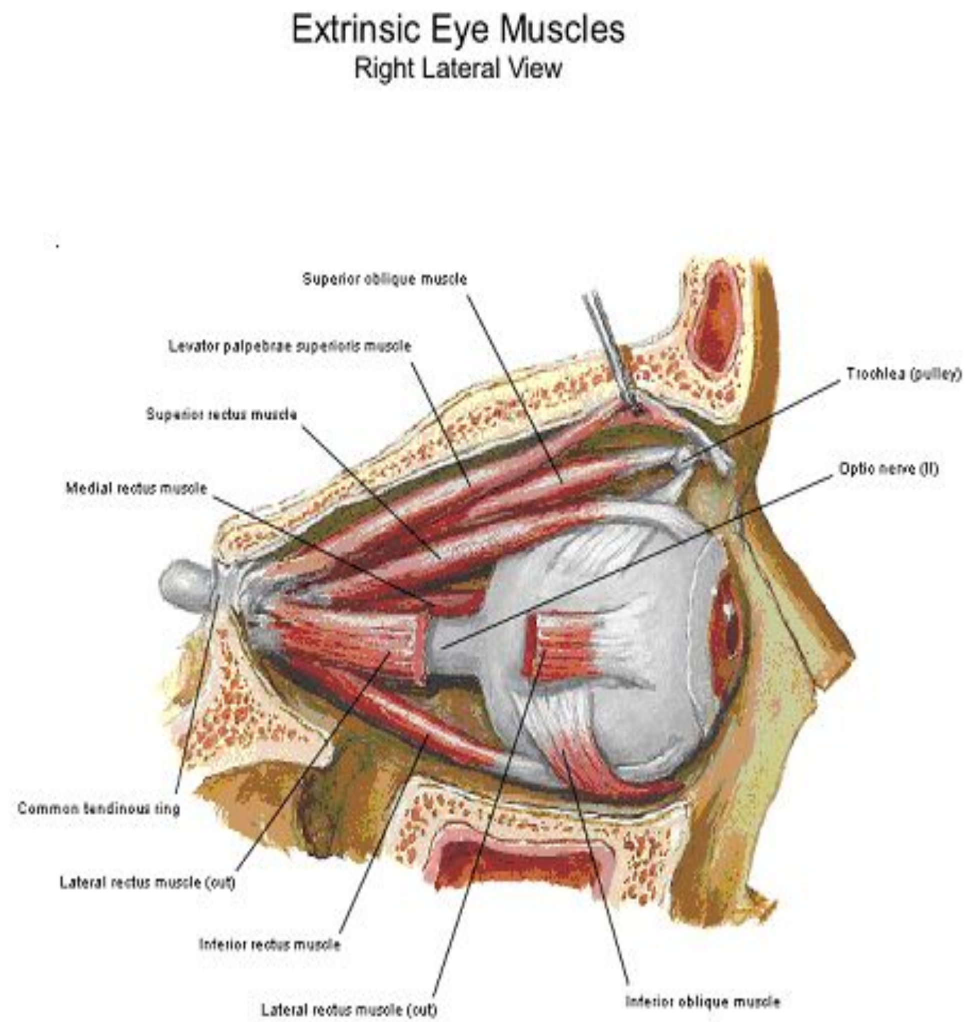
Extraocular Muscles

- Eyeball is moved by extrinsic or the extraocular muscles:

✔ Four recti

✔ Two obliqui

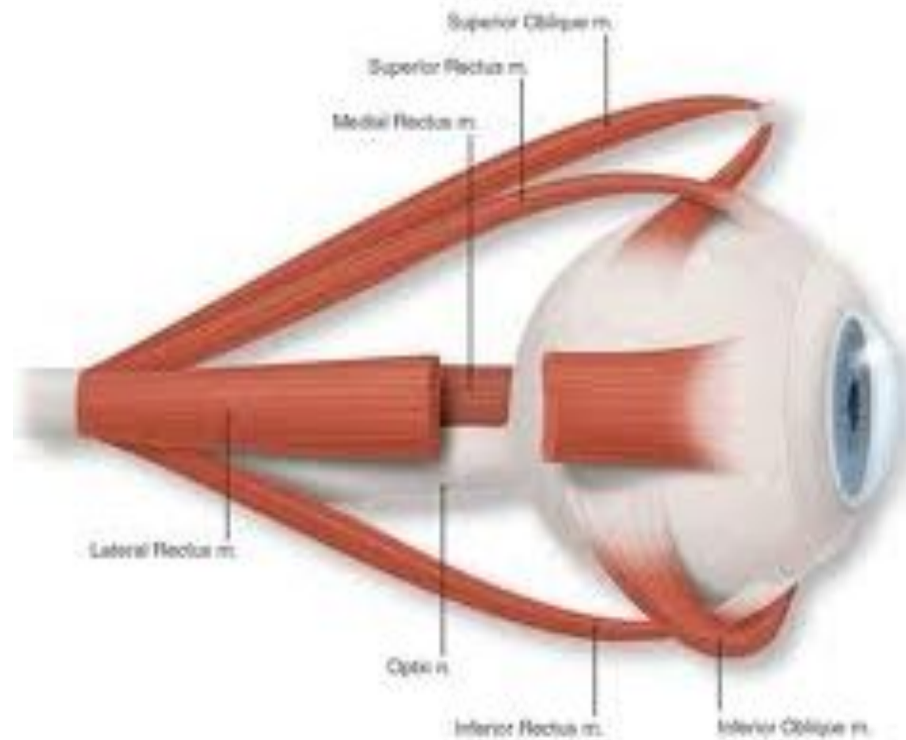
✔ Levator palpebrae superioris



THE RECTI MUSCLES

- The recti muscles arise from a common tendinous ring
- ✔ Superior & medial recti also arise from the dural sheath of the optic nerve
- ✔ Lateral rectus also arises from the orbital surface of the greater wing of the sphenoid bone
- ✔ All pierce the fascial sheath of the eyeball to get inserted into the sclera

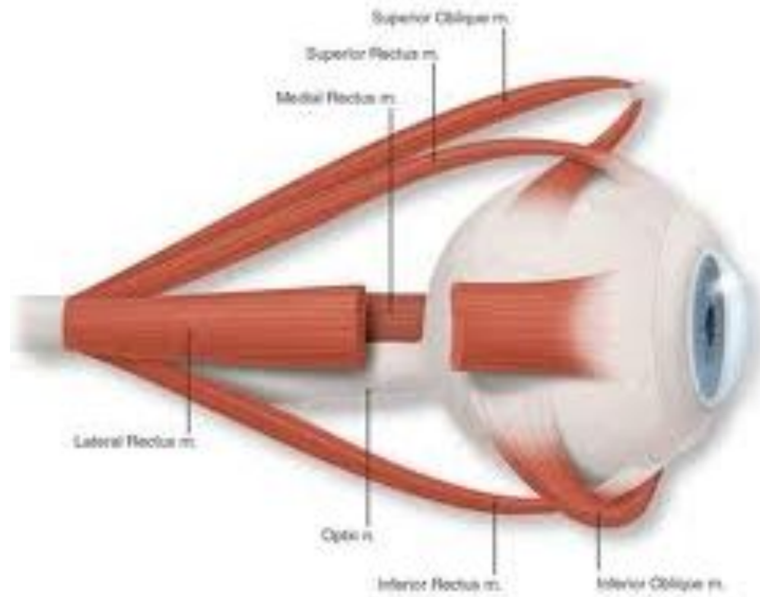
Superior oblique arises from body of the sphenoid and gets inserted into the posterolateral quadrant of the sclera



Inferior Oblique

Orbital surface of the maxilla (origin)

Posteroinferior quadrant of the sclera (insertion)



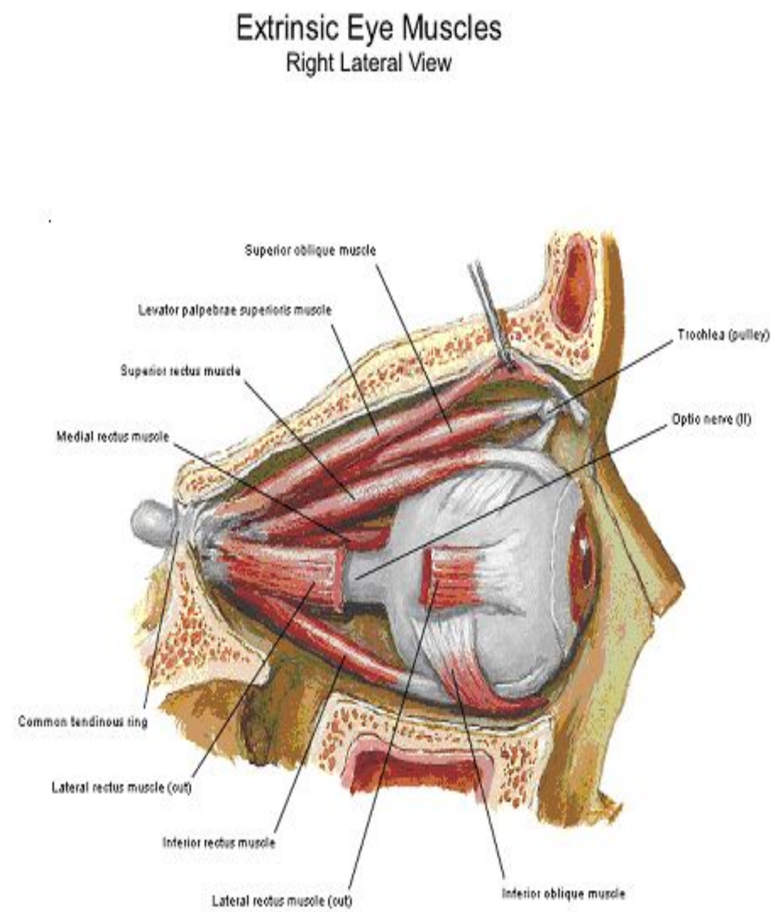
- **Levator palpabrae superioris**

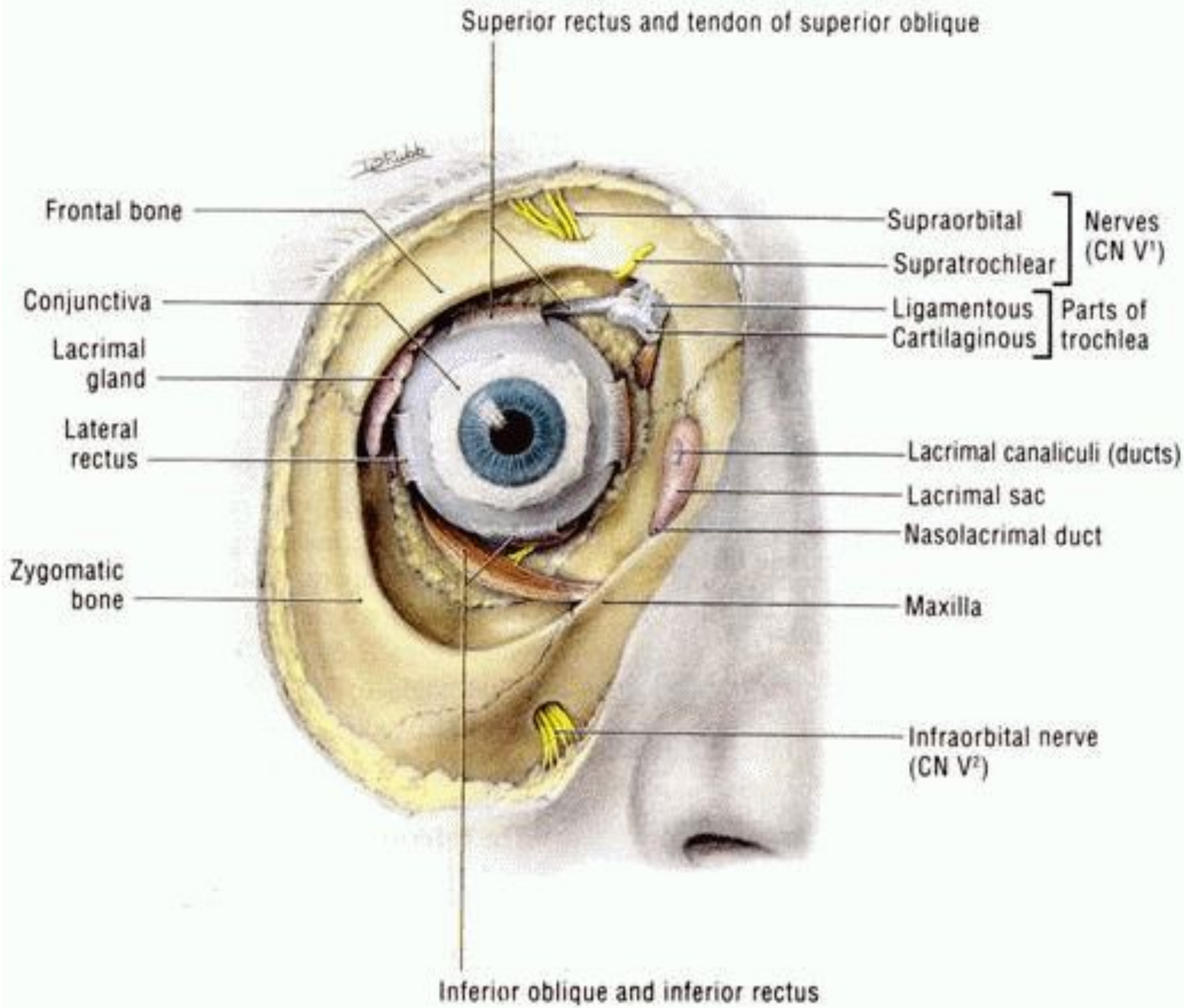
✔ arises from undersurface of the lesser wing of the sphenoid

✔ Splits into two lamellae

✔ Superior(voluntary) inserts into ant surface of superior tarsus and skin of upper eyelid

✔ Inferior(involuntary) inserts into upper margin of superior tarsus

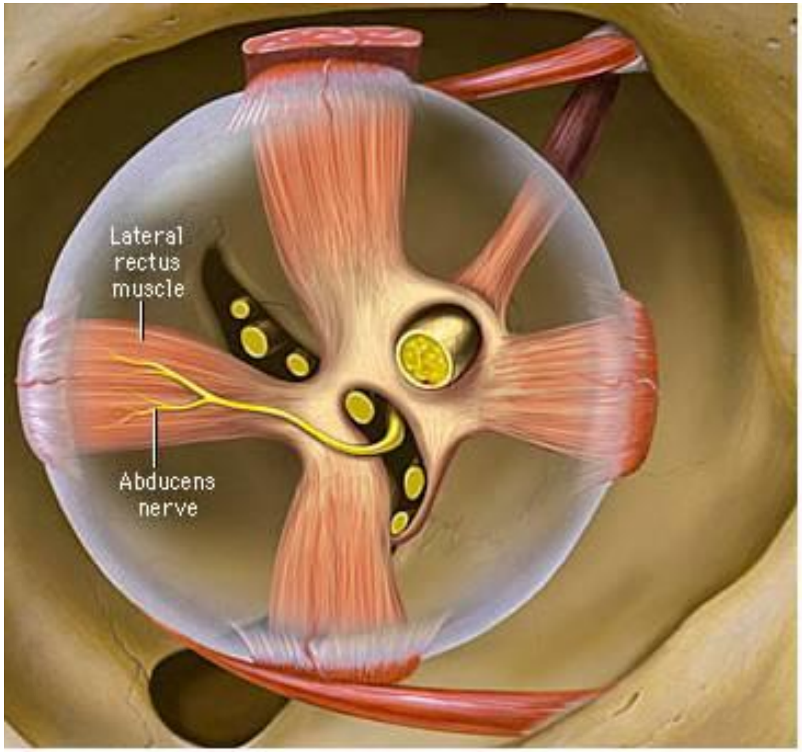




Superior rectus and tendon of superior oblique
Frontal bone
Conjunctiva
Lacrimal gland
Lateral rectus
Zygomatic bone
Supraorbital
Supratrochlear
Nerves (CN V¹)
Ligamentous
Cartilaginous
Parts of trochlea
Lacrimal canaliculi (ducts)
Lacrimal sac
Nasolacrimal duct
Maxilla
Infraorbital nerve (CN V²)
Inferior oblique and inferior rectus

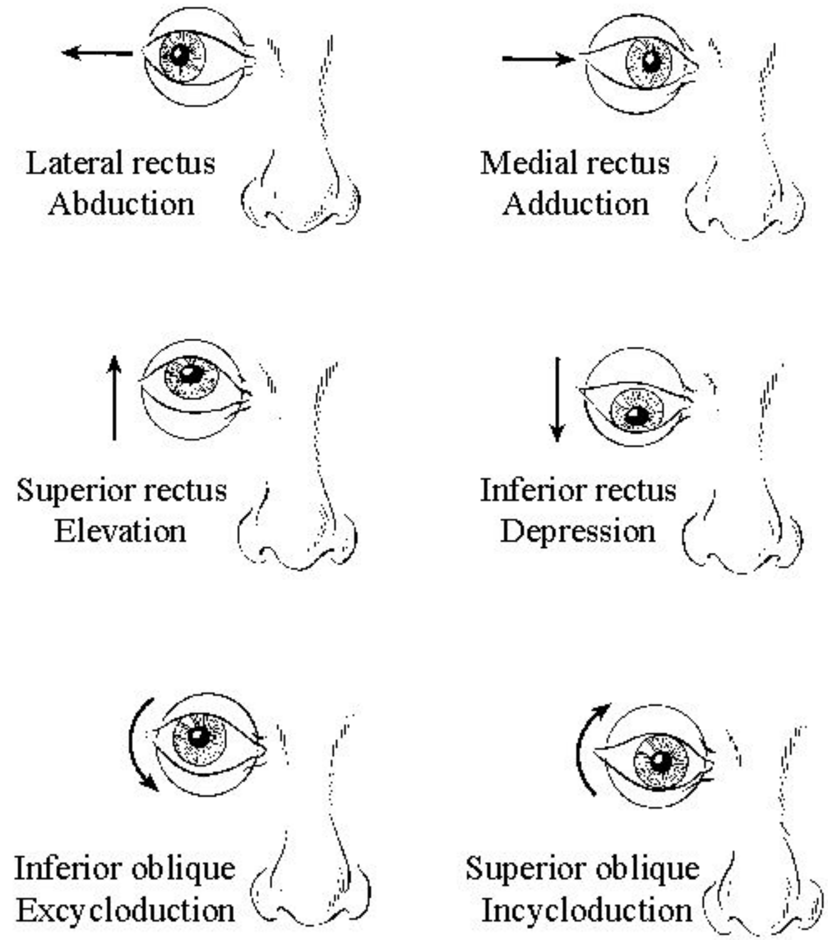
Nerve Supply

- (LR6 SO4)
- Lateral Rectus…abducent nerve
- Superior oblique…..Trochlear nerve
- Remaining muscles……occulomotor nerve

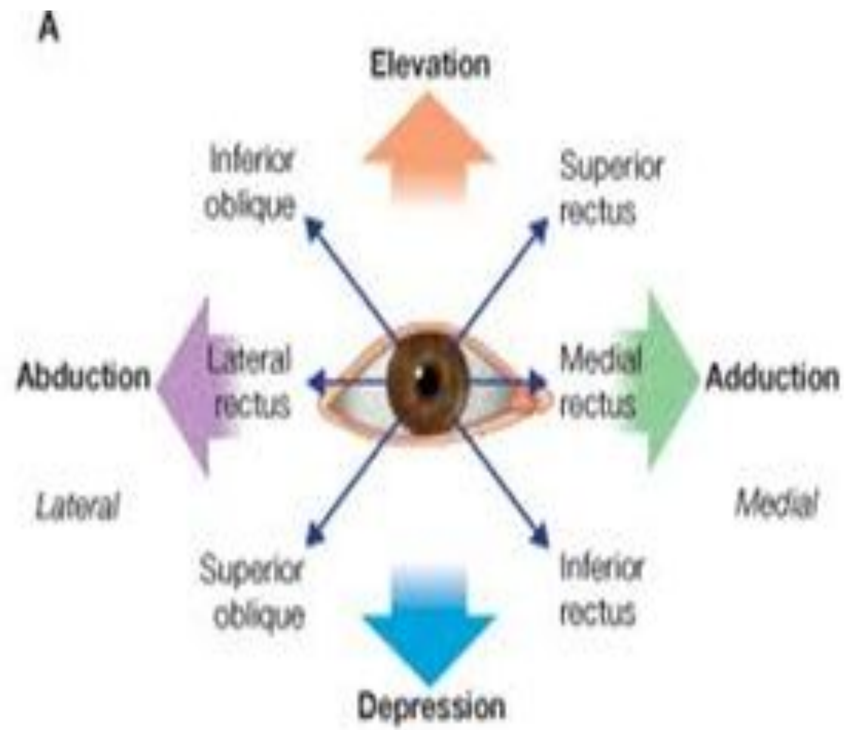


ACTIONS

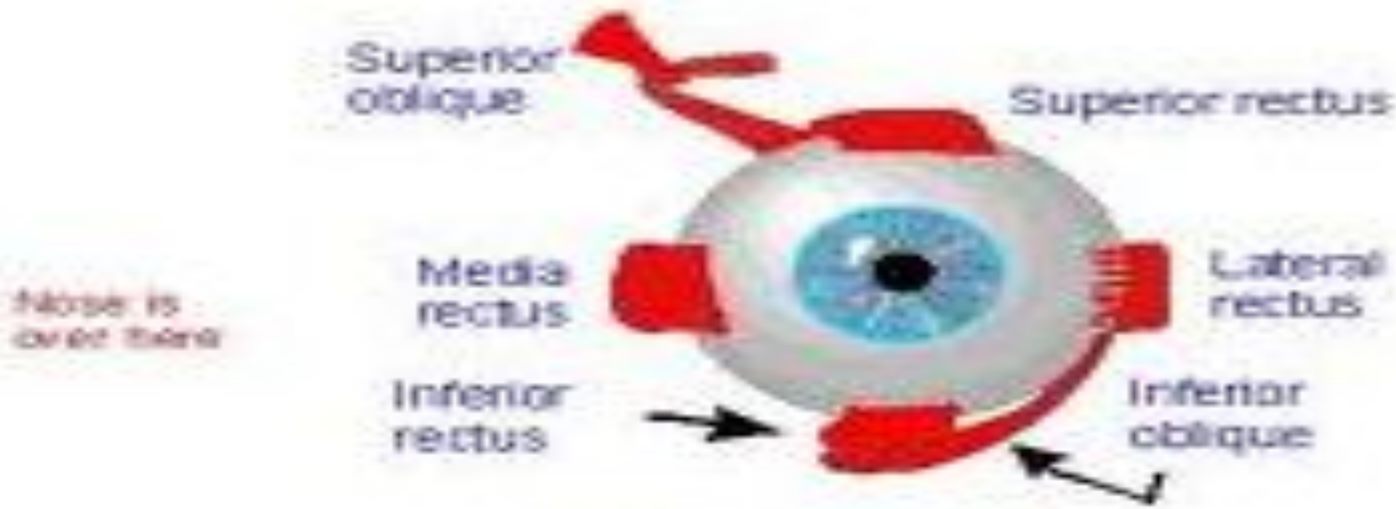
- Levator Palpabrae Superioris: Elevation of upper eyelid
- Superior rectus: Upward rotation + Medial rotation + Intortion
- Inferior rectus: Downward rotation + Medial rotation + Extortion



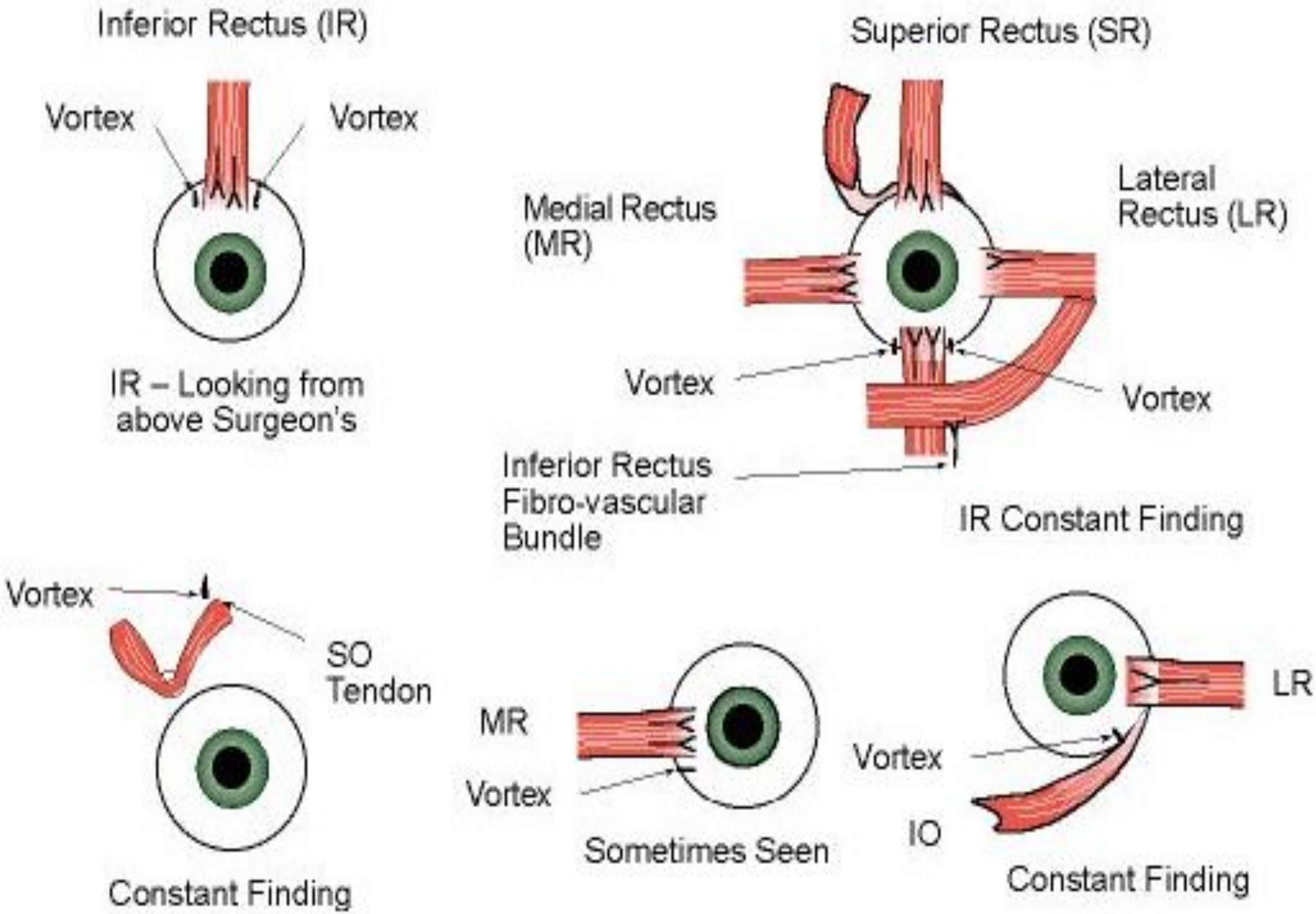
- **Medial rectus:**
  Medial rotation
- **Lateral rectus:**
  Lateral rotation
- **Superior oblique:**
  Downward rotation
  Lateral rotation
- **Inferior oblique:**
  Upward rotation
  Lateral rotation
  Extortion



- Medial rectus moves the eye towards the nose
- Lateral rectus moves the eye away from the nose
- Superior rectus moves the eye up
- Inferior rectus moves the eye down
- Superior oblique rotates the eye so that the top of the eye moves towards the nose
- Inferior oblique rotates the eye so that the top of the eye moves away from the nose



The Left Eye


Inferior Rectus (IR)
Vortex
Vortex
IR – Looking from above Surgeon's
Superior Rectus (SR)
Medial Rectus (MR)
Lateral Rectus (LR)
Vortex
Vortex
Inferior Rectus Fibro-vascular Bundle
IR Constant Finding
Vortex
SO Tendon
Constant Finding
MR
Vortex
Sometimes Seen
LR
Vortex
IO
Constant Finding